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# Practical No. 5
# Hash Table using Division Method and Linear Probing

SIZE = 10
hash_table = [None] * SIZE

# Insert a key
def insert(key):
    index = key % SIZE

    for i in range(SIZE):
        new_index = (index + i) % SIZE

        if hash_table[new_index] is None:
            hash_table[new_index] = key
            print("Key inserted successfully.")
            return

        if hash_table[new_index] == key:
            print("Key already exists.")
            return

    print("Hash table is full.")

# Search a key
def search(key):
    index = key % SIZE

    for i in range(SIZE):
        new_index = (index + i) % SIZE

        if hash_table[new_index] is None:
            break

        if hash_table[new_index] == key:
            print("Key found at index", new_index)
            return

    print("Key not found.")

# Delete a key
def delete(key):
    index = key % SIZE
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    for i in range(SIZE):
        new_index = (index + i) % SIZE

        if hash_table[new_index] is None:
            break

        if hash_table[new_index] == key:
            hash_table[new_index] = None
            print("Key deleted successfully.")
            return

    print("Key not found.")

# Display hash table
def display():
    print("\nHash Table:")
    for i in range(SIZE):
        print(i, ":", hash_table[i])

# Menu
while True:
    print("\n1. Insert")
    print("2. Search")
    print("3. Delete")
    print("4. Display")
    print("5. Exit")

    choice = input("Enter your choice: ")

    if choice == '1':
        key = int(input("Enter key: "))
        insert(key)

    elif choice == '2':
        key = int(input("Enter key to search: "))
        search(key)

    elif choice == '3':
        key = int(input("Enter key to delete: "))
        delete(key)

    elif choice == '4':
        display()
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    elif choice == '5':
        print("Program terminated.")
        break

    else:
        print("Invalid choice.")
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Sample Output :

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1. Insert
2. Search
3. Delete
4. Display
5. Exit
Enter your choice: 1
Enter key: 25
Key inserted successfully.

Enter your choice: 1
Enter key: 35
Key inserted successfully.

Enter your choice: 1
Enter key: 15
Key inserted successfully.

Enter your choice: 4

Hash Table:
0 : None
1 : None
2 : None
3 : None
4 : None
5 : 25
6 : 35
7 : 15
8 : None
9 : None

Enter your choice: 2
Enter key to search: 35
Key found at index 6

Enter your choice: 3
Enter key to delete: 35
```

Key deleted successfully.

Enter your choice: 4

Hash Table:
0 : None
1 : None
2 : None
3 : None
4 : None
5 : 25
6 : None
7 : 15
8 : None
9 : None

Enter your choice: 5
Program terminated.